

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-VI EXAMINATION – WINTER 2025****Subject Code:3150911****Date:02-12-2025****Subject Name:Power System- II****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Discuss Ferranti effect.	03
	(b) What is the p.u. system? Give the advantages of p.u. system	04
	(c) Explain Nominal-T and Nominal- π methods for analysis of medium transmission lines.	07
Q.2	(a) Give classification of transmission lines. Define voltage regulation and efficiency of a transmission line.	03
	(b) Discuss power in single phase AC circuits.	04
	(c) Explain per unit representation of transformers.	07
	OR	
	(c) Explain rigorous solution of long transmission line	07
Q.3	(a) What is the difference between symmetrical and unsymmetrical faults?	03
	(b) Explain Generalized Circuit Constant and derive ABCD Constant for short transmission lines.	04
	(c) Discuss transients on a transmission line in case of a symmetrical fault.	07
	OR	
Q.3	(a) What are the symmetrical components? Explain in short.	03
	(b) Prove that positive sequence and negative sequence impedance of fully transposed transmission line are always equal.	04
	(c) Write short note on an attenuation of travelling waves.	07
Q.4	(a) Explain in short Surge absorber	03
	(b) Explain the short circuit of a synchronous machine for no load condition.	04
	(c) Using appropriate interconnection of sequence networks, derive the equation for a line to line fault in a power system with fault impedance of Z_f .	07
	OR	
Q.4	(a) Show how Z_{bus} matrix gets modified when branch Z_b is added from new bus to old bus (type-2 modification)	03
	(b) Write a short note on selection of circuit breaker.	04
	(c) Explain phase shift in star-delta transformer.	07
Q.5	(a) Write a short note on the arcing ground phenomenon.	03
	(b) What is Corona? Explain different methods to reduce corona loss.	04
	(c) What is the harmful effect of lightning on a power system? Discuss the different types of lightning strokes.	07

OR

- Q.5** **(a)** What is Critical Disruptive Voltage? Write a formula of Critical Disruptive Voltage and briefly describe each variable in the formula. **03**
- (b)** Explain factor affecting corona, **04**
- (c)** What is the phenomenon of wave reflection and refraction? Derive expression for reflection and refraction. **07**
